Choose Better Hyperparameters: An extensive simulation study to aid the selection of optimal hyperparameters for Causal Discovery Algorithms

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Abstract :

Causal discovery algorithms have gained significant traction in various research domains, yet practitioners often lack intuitive understanding of critical tuning parameters that substantially impact results. This study addresses this gap by empirically investigating optimal parameter selection for score-based causal discovery approaches through comprehensive in-silico experimentation. We specifically focus on the penalty discount parameter, which controls the sparsity-complexity tradeoff in learned causal structures. Using a systematic evaluation framework across varying network configurations (20 and 100 variables with average degrees of 2 and 6), we analyze F1 scores, precision, and recall metrics at different sample sizes to identify optimal parameter values. Our results demonstrate that penalty discount values between 0.6-1.0 consistently yield superior F1 scores, with 1.0 providing an optimal balance between precision and recall for normally distributed data when using BOSS or similar BIC-optimizing algorithms with appropriate resampling methods. We observe that while higher penalty discounts (>2.0) favor precision, lower values prioritize recall, with this tradeoff diminishing at larger sample sizes. These findings provide evidence-based guidelines for parameter selection in score-based causal discovery algorithms, enabling more reliable structure learning in practical applications. The study contributes toward establishing transparent best practices that enhance both the accessibility and reliability of causal discovery methods for researchers across disciplines.

Index Terms: Hyperparameter, BIC, Causal Discovery.

1 Study Goals and Motivation

As causal discovery gains more and more popularity, there is a need of defining best practices for the parameters that are crucial to its use. Although the parameters have theoretical meaning, and definition, practitioners often lack the intuitive understanding of these parameters. In this paper, we include a varied range of datasets in an in-silico experiment, that might be used to define best practices that can be both comprehensible and lucid enough to enable intuitive understanding.
Causal Discovery is split into two approaches, a constraint based approach and a score based approach. In this paper, we investigate parameter tuning for methods involving score based approach only. Methods that involve score based approaches, have a penalty discount term. The general intuition in the domain is, higher the penalty discount, the emptier the graph. Conversely, lower penalty discount gives us a complete graph. The important thing to note is if the penalty discount is lowered too much, then the search starts fitting the noise around variables and starts treating them as effects. This can lead to erroneous edge discovery and therefore mis-interpretation of results. So, the optimal penalty discount, is low but not low enough that we fit noise and not high enough that we lose information. In an effort to solve this problem, this research work does an extensive sweep through different penalty discount values to identify the optimal penalty discount for a given dataset.

1.1 Contribution

The key contributions of the paper are:

- We conducted and present results from a simulation study of a grid search of hyperparameters (penalty discount) for algorithms (score-based) in the context of causal discovery, which is far more extensive than any previous simulation study of this topic.
- This study reveals via simulations that several penalty discounts are approximately the same, namely 0.8, 1.0, and 1.2 while using the Bootstrap adjusted with Effective Sample Size as a resampling technique.
- The optimum penalty discount parameter for linear gaussian distributed data with weak effects, is closer to 0.6-1.0 when paired with Bootstrap adjusted for sample size as a resampling method.
- The penalty discounts 0.6 had poor precision but had higher recall whereas the higher penalty discount 6.0 had close to perfect precision but had very poor recall.

2 Background

2.1 Directed Acyclic Graphs

A *directed acyclic graph* (DAG) is an edge vertex graph whose vertices and edges represent variables and conditional independence statements. We use the notation $\mathcal{G} = (V, E)$ where $\mathcal{G}$ is a graph, V is a vertex set, and E is an edge set. In a DAG, the edge set is comprised of ordered pairs that represent directed edges[1] and contains no directed cycles.

Let $\mathcal{G} = (V, E)$ be a DAG and $v \in V$:

$$\mathrm{pa}_{\mathcal{G}}(v) \equiv \{w \in V \ : \ v \leftarrow w \ \text{ in } \mathcal{G}\}$$
$$\mathrm{ch}_{\mathcal{G}}(v) \equiv \{w \in V \ : \ v \rightarrow w \ \text{ in } \mathcal{G}\}$$

are the *parents* and *children* of v.

A DAG encodes graphical conditional independence between disjoint subsets $A, B, C \subseteq V$ which can be read off of the DAG using *d-separation* and is denoted $A \perp\!\!\!\perp B \mid C \ [\mathcal{G}]$ (Dawid, 1979; Pearl, 1988; Lauritzen, 1996). This criterion admits the concept

[1] The edge $v \leftarrow w$ corresponds to the order pair (v, w).

of a *Markov equivalence class* (MEC) which—in the context of this paper—is a collection of DAGs that represent the same conditional independence relations (Spirtes et al., 1993).

2.2 *Causal Discovery*

DAGs are connected to causality by the *causal Markov* and *causal faithfulness* assumptions. A DAG is *causal* if it describes the true underlying causal process by placing a directed edge between a pair of variables if and only if the former directly causes the latter (Spirtes et al., 1993).

Causal Markov: If $\mathcal{G} = (V, E)$ is causal for a collection of random variables X indexed by V with probability distribution P, then for disjoint subsets $A, B, C \subseteq V$:

$$A \perp\!\!\!\perp B \mid C \; [\, \mathcal{G} \,] \quad \Rightarrow \quad A \perp\!\!\!\perp B \mid C \; [\, P \,].$$

More generally, this implication is called the *Markov property* and we say DAGs satisfying this property *contains* the distribution.

Causal faithfulness: If $\mathcal{G} = (V, E)$ is causal for a collection of random variables X indexed by V with probability distribution P, then for disjoint subsets $A, B, C \subseteq V$:

$$A \perp\!\!\!\perp B \mid C \; [\, P \,] \quad \Rightarrow \quad A \perp\!\!\!\perp B \mid C \; [\, \mathcal{G} \,].$$

Under the causal Markov and causal faithfulness assumptions, finding the MEC of the causal DAG is equivalent to identifying the simplest DAG(s) in terms of edge count that contains the data generating distribution. Causal discovery algorithms automate this search process.

2.3 *The Bayesian Information Criterion and Likelihood Ratio Test*

Causal Discovery algorithms are generally characterized as score-based or constraint-based. Both approaches assess conditional independence relationships in that data—score-based methods with a goodness-of-fit score and constraint-based methods with conditional independence tests. In this paper we focus on the Bayesian Information Criterion (BIC) (Schwarz, 1978; Haughton, 1988) and the Likelihood Ratio Test (LRT) (Anderson, 1984) for Gaussian DAG models.

For a Gaussian model characterized with DAG $\mathcal{G}$ the BIC on X is defined as:

$$\begin{aligned} \mathrm{BIC}(\mathcal{G}, X) &= \ell(\hat{\theta_{\mathcal{G}}}) - \frac{|E|}{2} \log(n) \\ &= \sum_{v \in V} \ell(\hat{\theta_{v|\mathrm{pa}_{\mathcal{G}}(v)}}) - \frac{|\mathrm{pa}_{\mathcal{G}}(v)|}{2} \log(n) \end{aligned}$$

where $\hat{\theta_*}$ is the MLE. Causal discovery methods generally comparing nested models that differ by a single edge. In what follows, let $\hat{\theta_{\nleftarrow}}$ be the MLE of the model without the edge and $\hat{\theta_{\leftarrow}}$ is the MLE of the model with the edge. When comparing two models that differ by a single edge the difference in BIC scores is computed as:

$$\Delta_{\mathrm{BIC}} = \ell(\hat{\theta_{\nleftarrow}}) - \ell(\hat{\theta_{\leftarrow}}) - \frac{1}{2} \log(n).$$

Similarly, the likelihood ratio test statistic for comparing two models that differ by a single edge is computed:

$$\lambda_{\mathrm{LR}} = -2 \left[\ell(\hat{\theta_{\nleftarrow}}) - \ell(\hat{\theta_{\leftarrow}}) \right]$$

where λ_{LR} is distributed χ^2_1. Noting that the MLE for a Gaussian distribution is computed as:

$$\ell(\hat{\theta}) = -\frac{n}{2} \left[\log(2\pi) + 2\log(\hat{\sigma}) - 1 \right]$$

we formulate the decision criterion for choosing between the two models (selecting the model with the edge) with the BIC as:

$$n \left[\log(\hat{\sigma_{\nleftarrow}}) - \log(\hat{\sigma_{\leftarrow}}) \right] - c\, \frac{1}{2} \log(n) > 0$$

where c is an extra term added to discount the parameter penalty. Similarly, we formulate the decision criterion for choosing between the two models (selecting the mode with the edge) with the LRT as:

$$2n \left[\log(\hat{\sigma_{\nleftarrow}}) - \log(\hat{\sigma_{\leftarrow}}) \right] - q > 0 \qquad \text{where} \quad \int_0^q \chi^2_1(x)\, dx = 1 - \alpha.$$

Accordingly, these decision criteria have the likelihood term (up to a multiplicative constant) and are tuned with c and α respectively.

2.4 *Metrics*

We use the following metrics to evaluate the learnt graphs by using an ensemble method to compare against the true data generating graph. The ensemble estimate is calculated by thresholding predicted probabilities at 0.5, i.e., considering no edge between two node pairs if the predicted probabilities are lesser than 0.5. For metrics that hint at calibration and reliability we calculate them via calculating an edge frequency table which stores the probabilites of two nodes being adjacent to each other. Then we compute the following metrics (Brier and ECE) using the edge frequency table.

2.4.1 Brier Scores In order to evaluate the calibration of the resampling techniques, we treat the resampling adjacency frequencies as predictive probability, and evaluate their predictive ability using the Brier score (Brier, 1950), a standard tool for evaluating the performance for binary events. The Brier score is simply the mean squared error between the predicted probability and the observed outcomes:

$$\text{Brier Score} = \frac{1}{n} \sum_{s=1}^{n} (f_s - o_s)^2 \qquad (1)$$

where n is the sample size, fs is the forecast probability of the event in sample s, and os is the observed outcome in sample s, with 0 meaning the event did not occur and 1 meaning the event did occur. Since this is a measure of error, larger values are worse: the best Brier score possible is 0, while the worst score possible is 1.

2.4.2 Expected Calibration Error

While Brier scores are commonly used to evaluate the calibration of probabilistic models, they provide an incomplete assessment of model calibration on their own. To gain a more comprehensive understanding, Expected Calibration Error (ECE) (Pakdaman Naeini et al., 2015) is often used in conjunction with Brier scores. ECE measures the discrepancy between predicted probabilities and observed outcomes by dividing the predictions into $\boldsymbol{K}$ equally spaced bins based on their confidence scores. Mathematically, ECE is defined as:

$$\text{ECE} = \sum_{i=1}^{K} P(i) \cdot |o_i - e_i| \tag{2}$$

where o_i is the true fraction of positive instances in bin i, e_i is the mean of the post-calibrated probabilities for the instances in bin i, and P(i) is the empirical probability (fraction) of all instances that fall into bin i. Lower ECE values indicate better calibration, as they suggest that the predicted probabilities more accurately reflect the observed frequencies of the positive class.

2.4.3 Precision, Recall, and F1-score

We report Precision, Recall, and F1-score to capture model performance (Sokolova et al., 2009). These metrics offer a more detailed understanding of how well a model identifies true positive instances while minimizing false positives and false negatives.

Precision measures the proportion of correctly predicted positive instances out of all instances predicted as positive. It is defined as:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{3}$$

where TP denotes true positives and FP denotes false positives. High precision indicates a low rate of false positives, ensuring that the positive predictions made by the model are reliable.

Recall, measures the proportion of actual positive instances that were correctly identified by the model. It is defined as:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{4}$$

where FN denotes false negatives. High recall ensures that most of the actual positive instances are captured by the model, minimizing missed detections.

To balance the trade-off between Precision and Recall, the F1-score is often used. The F1-score is the harmonic mean of Precision and Recall, providing a single composite metric that captures both aspects of model performance:

$$\text{F1 score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{5}$$

The F1-score ranges from 0 to 1, with higher values indicating better overall model performance. It is particularly useful in scenarios with class imbalance, where optimizing for either Precision or Recall alone may lead to suboptimal model behavior.

3 Data Description

3.1 Overview

To simulate the data we used the DaO method (Andrews et al., 2024) to simulate data involving two graph types, Erdos Renyi and Scale Free Graphs

The data were generated from a causal graphical model, followed by repeated resampling to create large collections of datasets. Each resampled dataset was analyzed using a causal discovery algorithm namely Best Order Score Search (BOSS) with a range of different hyperparameters (Penalty discount in this case), resulting in a corresponding large collection of graphs.

For each pair of variables, average degree, sample size, and the penalty discount, we calculated the proportion of times a specific relationship was represented by an edge (or lack of one) between them across the generated graphs. These proportions were then evaluated as predictive probabilities of whether an edge existed in the underlying data-generating model. We assessed the predictive probabilities (selecting a threshold of 0.5 for Precision, Recall, and F1 score) using several metrics, including the full Brier score, Expected Calibration Error (ECE), Precision, Recall and F1 score, comparing them to the true graphs that generated the data.

The entire process can be visualized by the picture below:

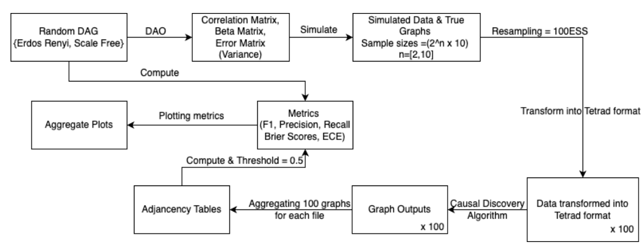

Figure 1. Simulation Flow Diagram

3.2 Data

To evaluate the performance of BOSS for different hyperparameter values, we performed an in-silico experiment involving simulations.

Random DAG: We generated random Erdős Rényi and scale free DAGs with $variables \in \{20, 100\}$ and $average\ degree \in \{2, 6\}$. For stability purposes, we generate 250 true graphs, for $variable = 20$, and 50 true graphs for $variable = 100$.

Correlation Matrix: The DAG is then used to compute a correlations matrix (and therefore a beta matrix and covariance matrix for additive error terms) using the DaO method (Andrews et al., 2024).

Simulated Data: Datasets are simulated using the beta matrix and covariance matrix for additive error terms. We generated datasets for several sample sizes such that $sample\ sizes = 2^n \cdot 10$ where $n \in \{2, \ldots, 10\}$.

4 Methods

The causal discovery algorithm was applied to each combination the simulation parameters as mentioned under the 3.2 using the Python wrapper of Tetrad (Joseph D Ramsey et al., 2018; Joseph D. Ramsey et al., 2023). The algorithm used in this paper is the Best

Order Score Search (BOSS) (Andrews et al., 2023). BOSS greedily searches over permutations of variables, using GSTs (Grow-Shrink Trees) to construct and score DAGs from permutations. GSTs efficiently cache scores to eliminate redundant calculations. BOSS achieves state-of-the-art performance in accuracy and execution time, comparing favorably to a variety of combinatorial and gradient-based learning algorithms under a broad range of conditions.

Causal discovery algorithms are often difficult to trust in real-world applications due to concerns about the reliability of their outputs. A commonly accepted strategy to increase confidence in the results is to evaluate the stability of the learned causal graphs under resampling. To assess the robustness of our findings, We use a resampling-based approach that applies the Bootstrap method, adjusted appropriately for the sample sizes. Since the primary objective of this project is to identify the optimal tuning parameter for score-based causal discovery methods, We perform a systematic sweep across a range of parameter values, from 0.6 to 6.0.

After the search completed, the graphs learnt for each iteration of the 100 resamples, were converted into edge frequency table averaging over the graphs on the basis of how frequently an edge existed in those 100 resampled graphs for each original file (50 for 100 variables and 250 for 20 variables). These frequency tables were then compared with the true graphs to compute metrics such as the Brier Score, Expected Calibration Error, F1 Score, Precision, and Recall with a threshold of 0.5.

4.1 Settings

In this study, a comprehensive grid search was conducted over a series of penalty discount values to determine whether resampling could serve as a reliable indicator of the optimal tuning parameter. Sixteen distinct penalty terms were evaluated, spanning a continuous range from 0.6 to 6.0. Specifically, the grid advanced in increments of 0.2 between 0.6 and 3.0, after which coarser increments of 1.0 were employed to cover the interval up to 6.0.

To assess the stability of the causal discovery outputs, the bootstrap method with adjusted sample sizes was implemented as the resampling technique. This approach allowed for repeated sampling with replacement, thereby providing insight into the variability of edge inclusion under different penalty regimes.

All algorithms were supplied with a precomputed correlation matrix, which is a mandatory input for the BOSS procedure. The correlation structure was estimated using the DaO method, ensuring that the matrix accurately reflected the underlying variable inter dependencies.

Model selection was driven by the structural equation model Bayesian Information Criterion (SEM-BIC) score, which was optimized at each iteration of the search algorithm. The SEM-BIC criterion balanced model fit and complexity, facilitating the identification of parsimonious causal structures.

The effective sample size was explicitly passed to the algorithm as required by the Tetrad framework, thereby ensuring that all computations adequately accounted for the number of observations. Although the number of random restarts (“starts”) is not formally treated as a tuning parameter within this work, it was maintained at the software default of one to preserve computational consistency across penalty settings.

4.2 Choices

In configuring the simulated data environments and algorithmic parameters for this study, the following decisions were made to ensure a comprehensive evaluation of causal discovery performance across diverse scenarios:

Graph Type, Variable Count, Average Degree, and Sample Size Two canonical network structures were selected to emulate both purely random and empirically motivated dependency patterns. The Erdős–Rényi (ER) model, characterized by n nodes and M uniformly random edges, served to benchmark performance on archetypal random graphs. In contrast, Scale-Free (SF) networks, which exhibit heavy-tailed degree distributions akin to many real-world systems, were included to assess algorithmic robustness under more heterogeneous connectivity. Variable counts were systematically varied to encompass both “thin” datasets (fewer variables) and “fat” datasets (greater dimensionality), while average degrees spanned from sparse to dense regimes. Sample sizes were likewise chosen to span small, moderate, and large cohorts, reflecting the range of dataset lengths commonly explored in causal discovery literature.

Algorithm Selection The BOSS (Bootstrapped Ordering Search over Score-based models) algorithm was employed due to its demonstrated ability to recover causal structure more efficiently and accurately than other score-based methods such as GES (Greedy Equivalence Search). BOSS’s search strategy adheres faithfully to the BIC criterion, thereby facilitating precise navigation of the variable space in pursuit of underlying causal relationships.

Penalty Discount and Number of Starts To strike a balance between over- and under-regularization, penalty discounts were assessed across a broad grid (as detailed in Section 3.1). Prior work by the author indicated that moderate penalties (e.g., values of 1.0 and 2.0) offer a suitable trade-off between false positives and false negatives. The number of random restarts (“starts”) was held at the default value of one, consistent with standard practice in instructional settings and to isolate the impact of the penalty parameter without introducing additional sources of variance.

Resampling Technique Informed by preliminary investigations Banerjee et al., 2025, the bootstrap with adjusted sample sizes was adopted as the resampling framework. Compared to the classic bootstrap, this variant yielded superior performance on key metrics—F1 score, precision, recall, and reliability—by preserving effective sample size and thereby enhancing the stability of edge-inclusion estimates. Formally, The idea of Bootstrap with effective/ adjusted sample size is explained as:

Let $X = (X_1, \ldots, X_n)$ be a sample of n i.i.d. instances and $\tilde{X}$ be a bootstrapped sample. While the number of instances in $\tilde{X}$ is still n, the effective sample size is smaller because of the sample contains repeated instances. To see this in practice, we consider a likelihood ratio test (LRT) on bootstrapped samples using the effective sample size and the original sample size in the LRT calculations. Figure 2 shows the distribution of the p-values from the likelihood ratio tests where the data were generated under the null hypothesis—two independent standard normal distributions. Notice that the p-values from the likelihood ratio tests are uniform when using the effective sample size. However, the p-values are clearly not uniform when using the original sample size.

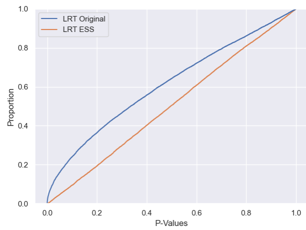

Figure 2. Non-uniformity of LRT p-values under the null hypothesis.

Let w_i be frequency weights (number of times the i^{th} instance appears in $\tilde{X}$). The effective sample size is calculated as follows (Kish, 1965):

$$n_{\text{ess}} = \frac{(\sum_{i=1}^{n} w_i)^2}{\sum_{i=1}^{n} w_i^2} = \frac{n^2}{\sum_{i=1}^{n} w_i^2} \approx \frac{n}{2}.$$

For a Gaussian model characterized with DAG $\mathcal{G}$ we calculate the BIC on $\tilde{X}$ with n_{ess} as follows:

$$\begin{aligned}
\text{BIC}(\mathcal{G}, \tilde{X}) &= n_{\text{ess}} \log(\hat{\theta_{\mathcal{G}}}) - c\,\frac{|E|}{2}\log(n_{\text{ess}}) \\
&\approx \frac{n}{2}\log(\hat{\theta_{\mathcal{G}}}) - c\,\frac{|E|}{2}\Big[\log(n) - \log(2)\Big] \\
&= \frac{n}{2}\log(\hat{\theta_{\mathcal{G}}}) - c\,\frac{|E|}{2}\log(n) + O(1) \\
&\propto n\log(\hat{\theta_{\mathcal{G}}}) - \tilde{c}\,\frac{|E|}{2}\log(n) + O(1)
\end{aligned}$$

where $\hat{\theta_{\mathcal{G}}}$ is the MLE and $\tilde{c} = 2c$. Accordingly, calculating the BIC on $\tilde{X}$ with n_{ess} is similar to using the original sample size and doubling the penalty discount term.

Scoring Criterion The SEM-BIC score was selected as the optimization target, reflecting both the linear Gaussian assumptions underlying the simulated datasets and the compatibility of the SEM-BIC criterion within the Tetrad framework. SEM-BIC's balance of model fit and complexity, combined with its prevalence in pedagogical examples, rendered it the natural choice for this investigation.

Implementation Details A high-performance C implementation of the BOSS algorithm, developed by Dr. Bryan Andrews, was utilized in place of the standard Java-based Tetrad version. This implementation, interfaced along JPype to access certain Tetrad features, afforded substantial gains in computational speed without compromising algorithmic fidelity.

Performance Metrics To quantify the accuracy of the recovered graph structures, we computed the F1 score alongside its constituent metrics of precision and recall. These metrics collectively capture the trade-offs between correctly identified edges and erroneous inclusions or omissions.

Adjacency Versus Edge-Frequency Analysis While adjacency matrices provide a binary indicator of inferred connections, more nuanced insights can be gleaned from edge-frequency tables generated across bootstrap iterations. Although implementation of frequency-based summaries proved technically challenging until recently, both binary adjacency and edge-frequency approaches are ultimately reported to offer a richer understanding of edge stability under resampling.

5 Results

Here are the Results:

Reporting a subset of results here, The exhaustive set of plots are not presented because of redundancy of information, however, they are available on request.

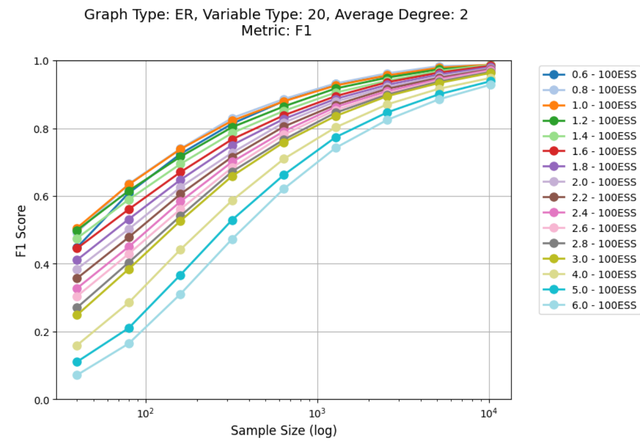

Figure 3. ER, 20 Variables, Average Degree 2. F1 plot

This graph displays F1 scores for an Erdős–Rényi (ER) network with 20 variables and an average degree of 2. The x-axis represents sample size on a logarithmic scale from 40 to 10,240, while the y-axis shows F1 scores ranging from 0 to 1. Multiple curves represent different penalty discount values (ranging from 0.6 to 6.0), each assessed after resampling with Bootstrap with effective sample sizes (100 ESS). The F1 scores generally improve as sample size increases across all parameter settings. Lower penalty discount values (0.6-1.4) achieve higher F1 scores, particularly at larger sample sizes, with values approaching 0.95-1.0 at the maximum sample size. Higher penalty discounts (4.0-6.0) perform notably worse at smaller sample sizes (0.1-0.2 at $n \approx 50$) but gradually improve with increased data this can be explained with them learning fewer but stable edges.

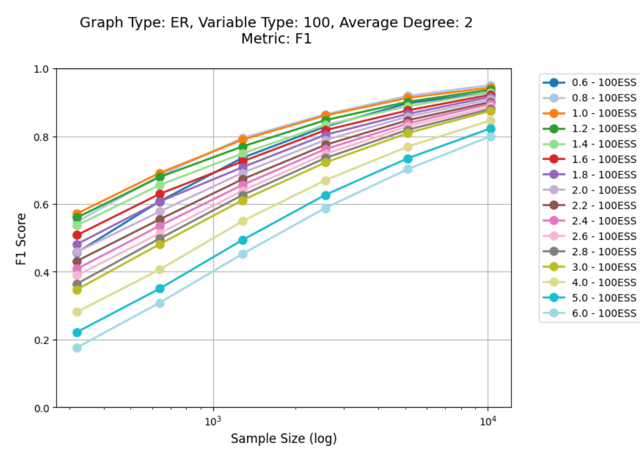

Figure 4. ER, 100 Variables, Average Degree 2. F1 plot

This graph examines F1 scores for a larger network with 100 variables and an average degree of 2. Despite the increased number of variables, the overall pattern remains similar, with F1 scores improving as sample size increases. At the lowest sample size, even the best-performing penalty discount parameters (0.8-1.2) achieve only moderate F1 scores (0.5-0.6). As sample size increases to 10,240, the best-performing parameters achieve F1 scores of approximately 0.95, while the highest penalty discount values (5.0-6.0) reach only about 0.8.

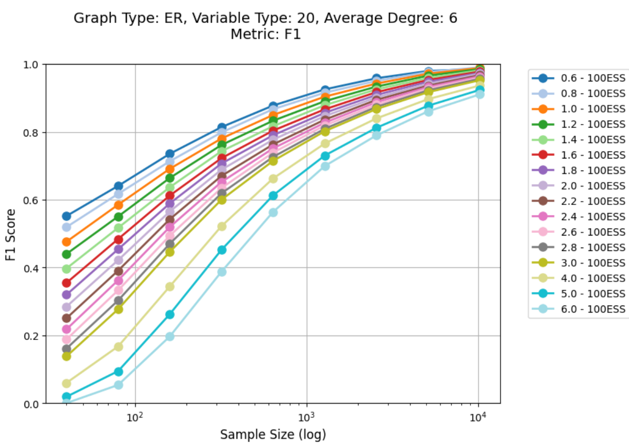

Figure 5. ER, 20 Variables, Average Degree 6. F1 plot

This figure presents F1 scores for an ER network with 20 variables but a higher average degree of 6. This is particularly interesting as the network is considerably dense compared to the previous graphs. The axes and parameters remain consistent with 3. A similar pattern of improvement with increasing sample size is observed. However, compared to the degree-2 network, there is a more pronounced stratification between penalty discounts, with lower values (0.6-1.0) demonstrating substantially better performance, particularly at mid-range sample sizes (100-1,000). At the maximum sample size, all parameter settings eventually converge toward high F1 scores (0.9-1.0), though the convergence rate differs significantly based on penalty discount values.

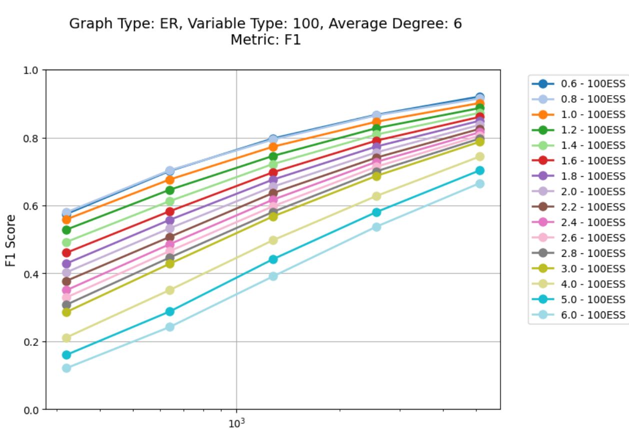

Figure 6. ER, 100 Variables, Average Degree 6. F1 plot

The final graph depicts F1 scores for an ER network with 100 variables and an average degree of 6. The sample size range matches 4. This configuration demonstrates the most challenging recovery scenario, with even the best-performing penalty parameters (0.6-0.8) achieving F1 scores of only 0.55-0.6 at the lowest sample size. The separation between different penalty terms is particularly pronounced in this high-dimensional connected network. At the maximum sample size of 10,240, the best penalty discounts reach F1 scores of approximately 0.9-0.95, while the highest penalty values (5.0-6.0) achieve only about 0.65-0.7.

Across all four network configurations, several consistent patterns emerge. F1 scores consistently improve with increasing sample size, demonstrating the expected relationship between data availability and structure recovery accuracy. Lower regularization values (0.6-1.4) generally outperform higher values, particularly in more complex networks and at smaller sample sizes. Network complexity significantly impacts recovery performance—as either the number of variables increases (from 20 to 100) or the average degree increases (from 2 to 6), higher sample sizes are required to achieve equivalent F1 scores. The convergence toward optimal F1 scores at large sample sizes varies based on network complexity, with simpler networks (20 variables, degree 2) approaching near-perfect recovery more rapidly than complex networks (100 variables, degree 6).

Although the F1 curves seem promising, here is an example of the two main components of the metric, Precision and Recall. Let's look at the 20 variables, average degree 2 case.

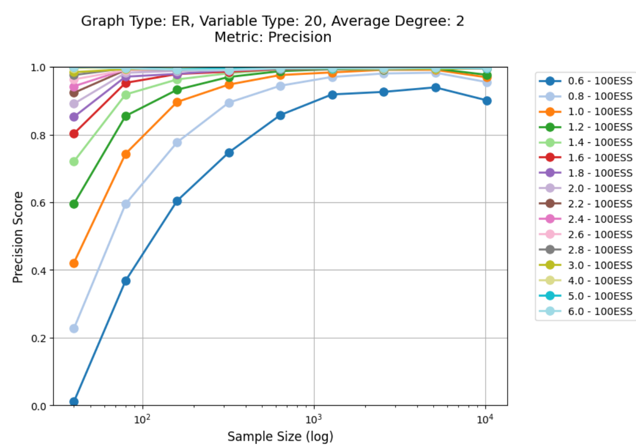

Figure 7. ER, 20 Variables, Average Degree 2. Precision plot

The precision scores exhibit distinct patterns across penalty discount values. Higher penalty discounts ($1.6 - 6.0$) demonstrate excellent precision (> 0.9) even at small sample sizes, maintaining consistently high values throughout because, the edges learnt are stable. In contrast, lower penalty discounts (0.6-1.0) show poor precision at small sample sizes (0.01-0.4 at $n \approx 40$) but improve rapidly as sample size increases, eventually converging to approximately 0.9-1.0 at larger sample sizes. Notably, the lowest penalty discount (0.6) never reaches the precision levels of higher values even at the maximum sample size, because, after a certain point the algorithm starts fitting noise instead of the actual causal signal.

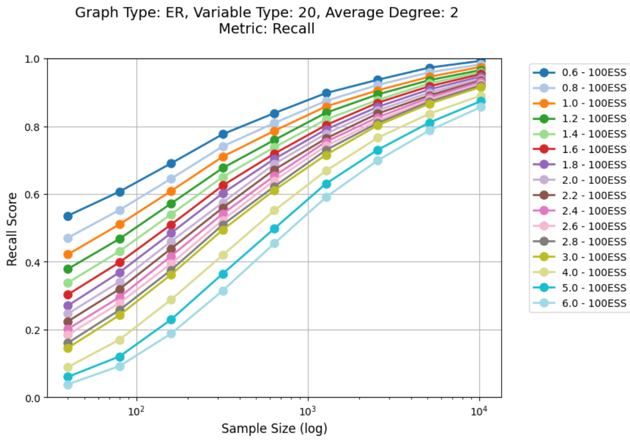

Figure 8. ER, 20 Variables, Average Degree 2. Recall plot

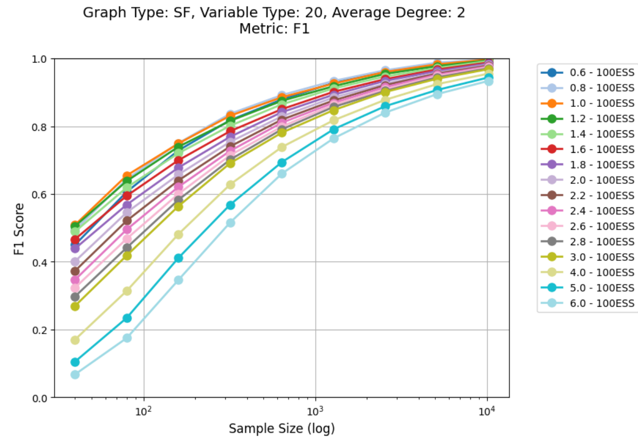

Figure 9. SF, 20 Variables, Average Degree 2. F1 plot

Unlike the precision results, recall scores demonstrate an inverse relationship with penalty discount values. Lower penalty discounts (0.6−1.0) achieve substantially better recall scores across all sample sizes, starting from moderate values (0.4 − 0.5) at the smallest sample size and increasing steadily to approximately 0.95-1.0 at the maximum sample size. Higher penalty discounts (4.0 − 6.0) perform poorly at small sample sizes (0.05 − 0.15 at $n \approx 50$) and improve more gradually, reaching only about 0.85−0.9 even at the largest sample size. All curves show consistent improvement with increasing sample size, though at different rates depending on the penalty discount value.

The precision and recall results for this network configuration reveal the fundamental trade-off inherent in structure learning approaches. Higher penalty discounts (1.6-6.0) strongly favor precision over recall, identifying fewer edges but with greater confidence. These values effectively control false positives but at the cost of increased false negatives, particularly at smaller sample sizes. Conversely, lower penalty discounts (0.6 − 1.0) prioritize recall over precision, identifying more true edges but also introducing more false positives, especially when data is limited. As sample size increases, both precision and recall improve across all penalty discount values, with the performance gap narrowing at the largest sample sizes. The optimal penalty discount appears to depend on the specific application requirements and available sample size, with moderate values (1.0-1.4) potentially offering the best balance between precision and recall, particularly at intermediate sample sizes.

Since the results for ER and SF graph types were almost identical, the focus in the results section is mainly on the ER networks. The graph type (ER or SF) did not significantly affect the outcomes, as both produced nearly identical results. This was also the case for the metrics such as F1, Precision, and Recall. The F1 plot for the Scale-Free (SF) graph type is shown below.

6 Discussion

The experimental results demonstrate several significant patterns regarding causal discovery algorithm performance across varying network configurations and sample sizes. These findings provide important insights for optimizing parameter selection in practical applications of structure learning algorithms.

Impact of Sample Size

The performance consistently improves with increasing sample size across all experimental conditions. The most substantial gains occur between small and moderate sample sizes (40-1,000 samples), with more gradual improvements thereafter. At 10,240 samples, most parameter configurations achieve high F1 scores (>0.9), demonstrating that sufficient data can compensate for suboptimal parameter selection. Precision stabilizes at smaller sample sizes than recall, particularly for higher penalty discount values, indicating that confidently identifying true edges requires less data than comprehensively recovering the complete network structure.

Effects of Network Complexity on Algorithm Performance

The comparative analysis of networks with different average degrees (2 versus 6) reveals nuanced effects of structural complexity on causal discovery performance. Networks with higher average degrees demonstrated a more pronounced stratification between penalty discount parameters at smaller sample sizes, with lower values (0.6-1.0) significantly outperforming higher values. Interestingly, despite these initial differences, the performance curves for different network densities converged toward similar F1 scores at the largest sample sizes, suggesting that the effect of network complexity on asymptotic performance is relatively modest. This finding aligns with theoretical expectations from the DaO simulations, which has primarily weak edges.

Influence of Variable Dimensionality on Baseline Performance

Counter to initial expectations, systems with higher dimensionality (100 variables) exhibited slightly higher baseline F1 scores (approximately 0.59) compared to lower-dimensional systems (20 variables, approximately 0.55) at moderate sample sizes. This phenomenon can be attributed to the increased sparsity of the learned graphs in high-dimensional settings. As the number of variables increases while maintaining the same average degree, the ratio of actual edges to possible edges decreases, enhancing the algorithm's ability to identify the absence of edges even with limited data. This observation has important implications for high-dimensional causal discovery, suggesting that while the absolute number of samples required for complete recovery increases with dimensionality, the relative performance at intermediate sample sizes may be more favorable than commonly assumed, particularly for sparse networks. This characteristic becomes especially relevant in domains such as genomics or neuroscience, where high-dimensional but sparse causal structures are prevalent.

Parameter Sensitivity and Optimization
Lower penalty discount values (0.6-1.0) consistently outperform higher values (>2.0) in F1 scores across all conditions. However, precision-recall analysis reveals that lower values prioritize recall at the expense of precision, while higher values show the opposite tendency. This trade-off is most pronounced at smaller sample sizes and in complex networks. Applications prioritizing precision should favor moderate to high penalty discount values (1.6-2.4), while those emphasizing comprehensive edge discovery benefit from lower values (0.6-1.0).

Implications for Practical Applications
Our findings suggest that adaptive parameter selection based on estimated graph properties and available sample size can significantly improve causal discovery performance. Resampling approaches effectively identify optimal parameter ranges for specific problem configurations, providing an alternative to theoretical parameter selection methods. The diminishing performance differences at larger sample sizes indicate that in data-rich environments, computational efficiency may be prioritized over parameter optimization. Conversely, in data-limited scenarios, careful parameter tuning becomes critical.

7 Conclusion and Future Work

7.1 TL;DR

For normally distributed data, when employing BOSS or other algorithms optimizing the Bayesian Information Criterion (BIC) score, our findings indicate that a penalty discount parameter of 1.0 provides optimal performance when using either bootstrap adjustment for sample size or 90% subsampling as resampling methods. This parameter value consistently yields superior structure recovery across various network configurations in terms of F1 score, particularly when balancing precision and recall metrics. It is important to note that this recommendation derives from simulations with weak effect sizes characteristic of DaO simulations. The optimal parameter value may differ in systems with substantial effect sizes among variables. Further research is warranted to establish parameter optimization guidelines for such contexts, as well as for non-Gaussian distributions and systems with latent confounders. This study contributes to the development of empirically-grounded best practices for causal discovery algorithm parameterization, facilitating more effective structure learning in applied research contexts.

7.2 Future work

Building upon the current findings, our research agenda encompasses two primary directions:

Extension to Empirical Data
We plan to validate and refine our parameter optimization framework using real-world datasets across multiple domains. This transition from simulation to empirical data will enable us to assess the generalizability of our findings and establish more robust parameter selection guidelines for practical applications. Additionally, this approach will allow us to evaluate causal discovery algorithm (CDA) performance under realistic conditions that may include violations of standard assumptions, irregular sampling distributions, and domain-specific constraints.

Distribution-Specific Parameter Optimization
We intend to develop a systematic methodology for transforming empirical data to approximate various statistical distributions (e.g., Gaussian, Poisson, exponential) and subsequently identify optimal parameter configurations for each distribution class. This distribution-specific approach aims to establish a comprehensive set of best practices that researchers can apply based on the statistical properties of their data. By mapping the relationship between data distributions and optimal tuning parameters, we seek to provide practitioners with clear guidance for parameter selection that maximizes structure recovery performance across diverse data types.

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